A COCONTINUOUS AND CONTINUOUS FUNCTOR COMMUTING WITH FIBRE PRODUCTS AND EQUALIZERS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Limits of sheaves are computed as presheaves. Same chapter, Lemma lemma-limit-sheaf. Identifies limits of sheaves with the corresponding limits of presheaves.
- Sheafification commutes with finite limits. Same chapter, Lemma lemma-sheafification-exact. Allows the reduction to the functor on presheaves.
- Colimits indexed by the auxiliary categories commute with finite limits. Same chapter, Lemma lemma-almost-directed and Categories, Lemma categories-lemma-almost-directed-commutes-equalizers. Supply the commutation of the colimits defining the left Kan extension with fibre products and equalizers.

Source reference. Original lemma 00W2: sources/stacks-sites/pinned-original-source.tex, source lines 4860–4860. Navigation pointer invoked at proof line 4860; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00WJ: sources/stacks-sites/pinned-original-source.tex, source lines 4863–4863. Navigation pointer invoked at proof line 4863; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original section 00VC: sources/stacks-sites/pinned-original-source.tex, source lines 4867–4867. Navigation pointer invoked at proof line 4867; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00X4: sources/stacks-sites/pinned-original-source.tex, source lines 4870–4870. Navigation pointer invoked at proof line 4870; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 002Y: sources/stacks-sites/pinned-original-source.tex, source lines 4872–4872. Navigation pointer invoked at proof line 4872; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ and $\mathcal{D}$ be sites. Let $u : \mathcal{C} \to \mathcal{D}$ be a functor. Assume that*

(a) *u is cocontinuous,*
(b) *u is continuous, and*
(c) *fibre products and equalizers exist in $\mathcal{C}$ and u commutes with them.*

In this case the functor $g_!$ above commutes with fibre products and equalizers (and more generally with finite connected limits).

Proof. Assume (a), (b), and (c). We have $g_! = (u_p\ )^\#$. Recall (Lemma 00W2) that limits of sheaves are equal to the corresponding limits as presheaves. And sheafification commutes with finite limits (Lemma 00WJ). Thus it suffices to show that u_p commutes with fibre products and equalizers. To do this it suffices that colimits over the categories $(\mathcal{I}_V^u)^{opp}$ of Section 00VC commute with fibre products and equalizers. This follows from Lemma 00X4 and Categories, Lemma 002Y. □